ABCPROOF · MOTIZUKIPDF · FIELD GUIDE NO. 2

Anabelian Reconstruction Algorithms

How inter-universal Teichmüller theory rebuilds fields, curves and functions from nothing but a group （遠アーベル幾何の復元アルゴリズム入門）



Sources: the papers in `pdf/` · numbers from `src/iut` and from the worked examples in `scripts/anabelian/build.js` · November 2026

ABSTRACT Anabelian geometry asks how much of a variety you can recover from its étale fundamental group alone. Mochizuki's *mono-anabelian* version asks for more than a yes: it asks for an *algorithm* whose only input is one abstract profinite group, and whose output is the field, the curve and its functions, with addition and multiplication. Inter-universal Teichmüller theory (IUT) needs these algorithms because its Θ-link glues two universes along groups and monoids but never along rings, so the only way to talk about the ring on the far side is to rebuild it from the shared group. This guide walks through the algorithms one at a time: the local invariants of a Galois group, Uchida's addition-from-multiplication lemma, cyclotomes and Kummer theory, Belyi and elliptic cuspidalization, the full reconstruction theorem [AbsTopIII, Thm 1.9], the archimedean version, the étale-like / Frobenius-like dichotomy and mono-anabelian transport, the three kinds of cyclotomic rigidity, and log-shells. Each one comes with a picture, a worked example whose numbers are computed by the build script, an analogy, and at least one joke. No mathematical claim rests on a joke.

étale-like: groups and containers (“zero mass”) · Frobenius-like: monoids, Frobenioids (“positive mass”) · reconstructed rings and fields · the algorithm · links and indeterminacies

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CONTENT NOTICE

All pop-culture references are text only: no character art, logos or screenshots. Musubi, the onigiri detective in the figures, is an original drawing. Where we say what a theorem states, we cite the paper and the numbered result. The 2018 disagreement about IUT is about [IUTchIII] Corollary 3.12, which comes after everything in this guide; we report it in §14 and adjudicate nothing. 審判ではありません。

0 Why IUT needs reconstruction algorithms なぜ復元が必要か

Here is the problem in its smallest form, taken from Mochizuki's 2023 Oberwolfach report [Gateway, §2]. Let R be an integral domain, say $\mathbb{Z}$, acted on by a group G, and let $N \geq 2$. The N-th power map $x \mapsto x^N$ is an isomorphism of multiplicative monoids from $R^{\triangleright} = R \setminus \{0\}$ onto its image, and it respects the G-action, but it is not a ring map, because $(x+y)^N \neq x^N + y^N$. Now take two labelled copies ${}^{\dagger}R$ and ${}^{\ddagger}R$ and glue them along it:

$$ {}^{\dagger}G \curvearrowright {}^{\dagger}R \;\supseteq\; ({}^{\dagger}R^{\triangleright})^N \;\xleftarrow{\sim}\; {}^{\ddagger}R^{\triangleright} \;\subseteq\; {}^{\ddagger}R \curvearrowleft {}^{\ddagger}G. $$

The groups ${}^{\dagger}G$ and ${}^{\ddagger}G$ can be identified, and so can the monoids, but the rings cannot. That is the shape of IUT's **Θ-link**: it identifies the Θ-pilot $\{q_v^{j^2}\}$ on one side with the q-pilot $\{q_v\}$ on the other, multiplicatively, and the only things it lets through are abstract groups (the Galois groups G_v) and abstract monoids. If you want to say anything about the ring structure of ${}^{\dagger}R$ in terms of ${}^{\ddagger}R$, you need a procedure that starts from the shared group and rebuilds as much ring structure as possible on both sides *by the same recipe*. A procedure whose input is a bare group and whose output is a field is exactly what anabelian geometry, in its algorithmic form, provides.

Mochizuki puts the connection to abc in one sentence [Gateway, §4; Alien, §3.11 (iii)]: anabelian geometry is about reconstructing *addition* from *multiplication* (plus topology), and the abc inequality bounds the additive size of a number (its height) by its multiplicative size (its conductor). Both say that addition is dominated by multiplication. The rest of this guide is about how the reconstruction is actually done.

> **OTAKU NOTE: THE PREMISE OF THE SERIES**
> Two worlds, one shared object that passes between them, and the rule that you may not bring anything else across. It is the setup of every isekai and of every "switched bodies" film: in one famous one, the two leads cannot even keep each other's names, only the notes they leave. IUT's notes are Galois groups. And since the Japanese word 結び (*musubi*) means both "rice ball" and "a knot that ties two things together", our mascot is an onigiri.

1 What anabelian geometry is 遠アーベル幾何とは

Every connected scheme X has an étale fundamental group $\pi_1(X)$, a profinite group that classifies its finite étale covers the way the topological fundamental group classifies covering spaces. For a geometrically connected variety over a field k there is a fundamental exact sequence

$$ 1 \longrightarrow \Delta_X \longrightarrow \Pi_X \longrightarrow G_k \longrightarrow 1, \qquad \Pi_X = \pi_1(X),\;\; \Delta_X = \pi_1(X_{\overline{k}}),\;\; G_k = \mathrm{Gal}(\overline{k}/k). $$

Δ_X is the *geometric* part (for a curve of genus g with r punctures it is the profinite completion of the topological fundamental group of a surface) and G_k is the *arithmetic* part. In his 1983 letter to Faltings, Grothendieck proposed that for "anabelian" varieties, whose fundamental groups are very far from abelian, the group Π_X together with its map to G_k should determine X. The prototypes are:

- **Number fields** (Neukirch, Uchida, 1969–1976): an isomorphism $G_F \cong G_{F'}$ of absolute Galois groups of number fields comes from a unique isomorphism $F' \cong F$.
- **Hyperbolic curves** (Tamagawa 1997 for affine curves; Mochizuki 1996 and 1999 for all hyperbolic curves over sub-p-adic fields, i.e. subfields of finitely generated extensions of $\mathbb{Q}_p$): isomorphisms of Π_X over G_k come from isomorphisms of curves [pGC, LocAn].

- **Absolute** results, where G_k is not given in advance and must itself be found inside Π_X: [AbsAnab], [AbsTopI–III]. These are the ones IUT uses.

A curve is *hyperbolic* when $2g - 2 + r > 0$. That condition is what makes Δ_X nonabelian and "rigid": $\mathbb{P}^1$ minus three points (a *tripod*) and an elliptic curve minus its origin are the two smallest examples, and both are the stars of this guide.

object	what it is	toolkit identifier	analogy
Π_X	arithmetic fundamental group of X; in IUT Π_v, tempered at bad v	`PiV`, `PiTpG`	the whole shadow
Δ_X	geometric fundamental group, kernel of $\Pi_X \twoheadrightarrow G_k$	`DeltaV`	the shape of the shadow
G_k	absolute Galois group of the base field	`GV`	the light source
$I_x \subseteq D_x$	inertia and decomposition groups of a point or cusp x	`InertiaEpsilonPrime`	the footprint at x
$\widehat{\mathbb{Z}}(1)$, M_X	a cyclotome: a copy of $\varprojlim \mu_N$; $M_X = \mathrm{Hom}(H^2(\Delta_X, \widehat{\mathbb{Z}}), \widehat{\mathbb{Z}})$	`MXCyclotome`	the tuning fork
Kummer map κ	$f \mapsto$ the class in $H^1(\Pi, \widehat{\mathbb{Z}}(1))$ of the torsor of roots of f	`KummerReady`	a transformation belt
$\mathcal{B}(\Pi)$	the anabelioid of finite Π-sets: "a space that is only its covers"	`BPi`, `Temperoid`	the Galois category

The toolkit's catalogue (`iut ids`) has 448 identifiers, 37 of kind *group* and 5 of kind *anabelioid*, and 73 processes. None of them computes with profinite groups; they model the data and its bookkeeping (§14).

OTAKU NOTE: WHO'S THAT CURVE?
Every Saturday morning a children's show from the late 1990s shows a black silhouette and asks you to name the creature. Anabelian geometry is that segment for algebraic geometry: the fundamental group is the silhouette, and the theorem says that for hyperbolic curves the silhouette is never ambiguous. The abelian case is the reason for the name: an elliptic curve's silhouette is just $\widehat{\mathbb{Z}}^2$, which every elliptic curve has, so the game is no fun. Ana-bel-ian = *not* that.

2 Mono-anabelian versus bi-anabelian 単遠アーベル と 双遠アーベル

There are two ways to say "the group determines the curve". The classical way, which Mochizuki calls **bi-anabelian**, compares two objects: given X_1, X_2 and an isomorphism $\Pi_{X_1} \cong \Pi_{X_2}$, it says the isomorphism comes from geometry. The **mono-anabelian** way is software: an algorithm that takes as input *one* abstract profinite group Π, which "just happens" to be isomorphic to some Π_X, and outputs field-like and scheme-like structures, functorially in Π [AbsTopIII, §I2, Rem 1.9.8]. The algorithm must be written "in language that only depends on the topological group structure": no peeking at how Π was built.

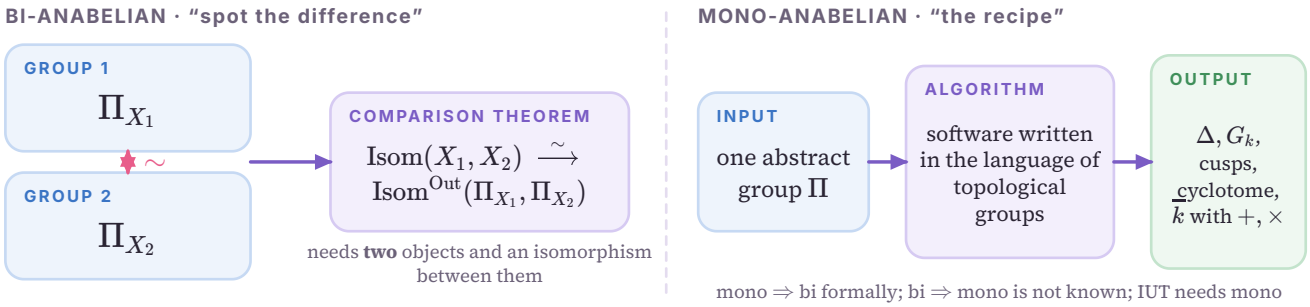


Figure 1. Bi-anabelian results compare two groups; mono-anabelian results are algorithms on one. A functorial algorithm gives a comparison theorem for free, so mono ⇒ bi. The converse is not known: an "algorithm" allowed to compare with a fixed reference model is really a bi-anabelian result in disguise [AbsTopIII, Rem 1.9.8].

Why IUT insists on mono: on the far side of a Θ-link (or a log-link) there is no reference model. The only thing you are handed is the group G_v or Π_v "as an abstract topological group", and anything you build from it must be built the same way in both universes. [AbsTopIII, §3] makes this precise: the mono-anabelian algorithms are compatible with the log-Frobenius operation (Cor 3.6), and the bi-anabelian approach is not (Cor 3.7 and the remarks after it). Mochizuki also remarks that the Neukirch–Uchida theorem was historically proved bi-anabelianly, and that [AbsTopIII, Thm 1.9] gave the first independent proof with an explicit construction of the number field [AbsTopIII, Rem 1.9.5]. Hoshi later gave a mono-anabelian reconstruction of a number field from its absolute Galois group alone [Hoshi19], and wrote the standard introduction to the subject [Hoshi21].

OTAKU NOTE: SPOT THE DIFFERENCE VS. THE VENDING MACHINE
Bi-anabelian geometry is the "find the 7 differences" page of a weekend newspaper: two pictures side by side, and the theorem says there are zero differences. Mono-anabelian geometry is a Japanese vending machine: you put in one coin (a profinite group), press one button (the algorithm), and a hot can of corn soup (a field with $+$ and $\times$) drops out. The machine has never seen your coin before and does not ask where you got it.

3 Algorithm A: what G_k knows about a p-adic field 局所体のガロア群

Start with the smallest input: $G_k = \mathrm{Gal}(\overline{k}/k)$ for a finite extension k of $\mathbb{Q}_p$ (an **MLF**, mixed-characteristic local field). Local class field theory says $G_k^{\mathrm{ab}} \cong (k^\times)^\wedge \cong \widehat{\mathbb{Z}} \times \mu_{q-1} \times \mu_{p^\infty}(k) \times \mathbb{Z}_p^d$, where $q = p^f$ is the size of the residue field and $d = [k : \mathbb{Q}_p] = ef$. That is the key to the following algorithm, which is [AbsAnab, Prop 1.2.1] read as software.

ALGORITHM A · INPUT: AN ABSTRACT PROFINITE GROUP $G \cong G_K$

1. Compute G^{ab} and its rank over $\mathbb{Z}_l$ for each prime l. It is 1 for $l \neq p$ and $d+1 \geq 2$ for $l = p$. So p is the unique prime with rank at least 2, and $d = [k : \mathbb{Q}_p]$ is that rank minus one.
2. The prime-to-p torsion of G^{ab} has order $q-1$. So q, $f = \log_p q$ and $e = d/f$ are determined.
3. An open subgroup H corresponds to an unramified extension exactly when e does not grow; the intersection of those is the **inertia subgroup** I_k.
4. The **Frobenius element** of $G/I_k \cong \widehat{\mathbb{Z}}$ is the one that acts on the prime-to-p roots of unity of unramified H as $x \mapsto x^q$.
5. The images of $\mathcal{O}_k^\times$, $k^\times$ in G^{ab} follow, and taking the torsion of the abelianizations of all open H, glued by the transfer (Verlagerung), gives a copy $\mu_{\mathbb{Q}/\mathbb{Z}}(G)$ of the roots of unity with its G-action: the **cyclotomic character**.
6. Finally the Brauer-group residue map $H^2(G, \mu_{\mathbb{Q}/\mathbb{Z}}(G)) \cong \mathbb{Q}/\mathbb{Z}$ of local class field theory is group-theoretic too [AbsAnab, Prop 1.2.1 (vii); AbsTopIII, Cor 1.10 (i)].

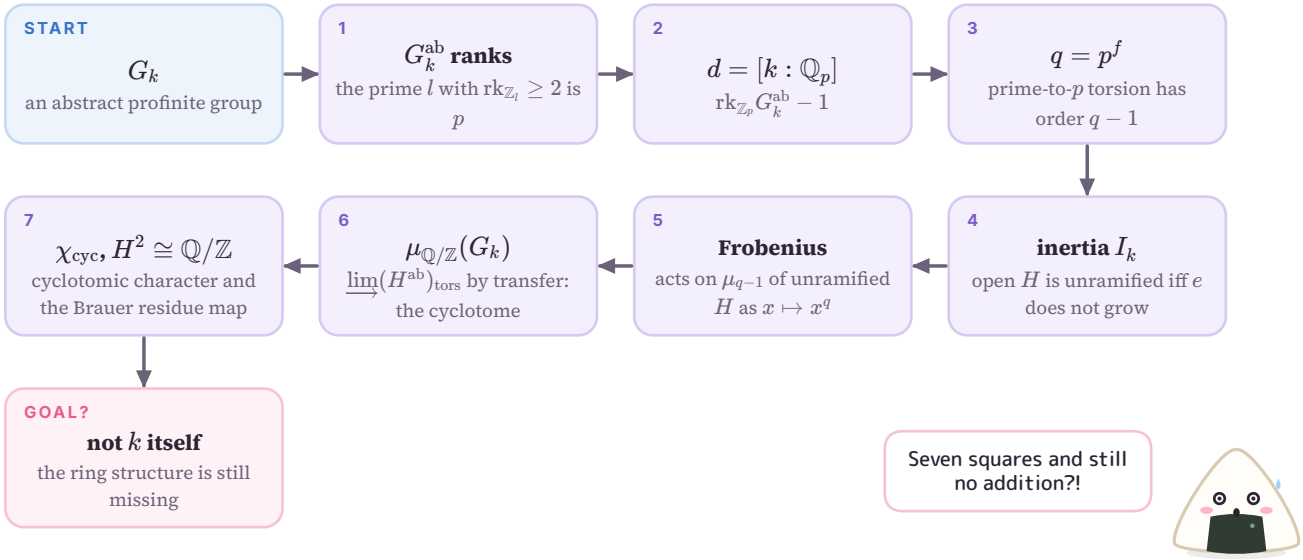


Figure 2. Algorithm A as a sugoroku board. Every square is something an arbitrary automorphism of G_k must preserve [AbsAnab, Prop 1.2.1]. The last square is the catch.

Here is what the algorithm reads off for six fields. The ranks are listed for $l = 2, 3, 5, 7, 11, 13$, with the rank at p in bold; the build script recomputes p, d, q, f, e from those numbers and stops if any comes out wrong. The last column is the order of the p-power roots of unity in k.

k	$\mathrm{rk}_{\mathbb{Z}_l} G_k^{\mathrm{ab}}$, $l = 2, \ldots, 13$	p	d	$q-1$	q	f	e	$\lvert\mu_{p^\infty}\rvert$	note
$\mathbb{Q}_3$	1 **2** 1 1 1 1	3	1	2	3	1	1	1	the bad place of $(1, 8, 9)$
$\mathbb{Q}_7$	1 1 1 **2** 1 1	7	1	6	7	1	1	1	$l = 7$ for $(1, 8, 9)$
$\mathbb{Q}_9$	1 **3** 1 1 1 1	3	2	8	9	2	1	1	unramified, degree 2
$\mathbb{Q}_3(\zeta_3)$	1 **3** 1 1 1 1	3	2	2	3	1	2	3	ramified, has μ_3
$\mathbb{Q}_3(\sqrt{3})$	1 **3** 1 1 1 1	3	2	2	3	1	2	1	ramified, no μ_3
$\mathbb{Q}_2(i)$	**3** 1 1 1 1 1	2	2	1	2	1	2	4	has μ_4

Look at the rows for $\mathbb{Q}_3(\zeta_3)$ and $\mathbb{Q}_3(\sqrt{3})$ have the same p, d, q, f, e, and Algorithm A only tells them apart through the p-power torsion. In general it cannot do more. There exist non-isomorphic p-adic

fields with isomorphic absolute Galois groups (Yamagata, Jarden–Ritter; see [NSW], the closing remark before Thm 12.2.7), and Mochizuki's 1997 paper records that he set out to prove the naive Grothendieck conjecture for p-adic fields and found it to be false [GCpLF, §0]. What *is* true is that G_k together with its higher ramification filtration determines k [GCpLF, Thm 4.2], a result proved with p-adic Hodge theory.

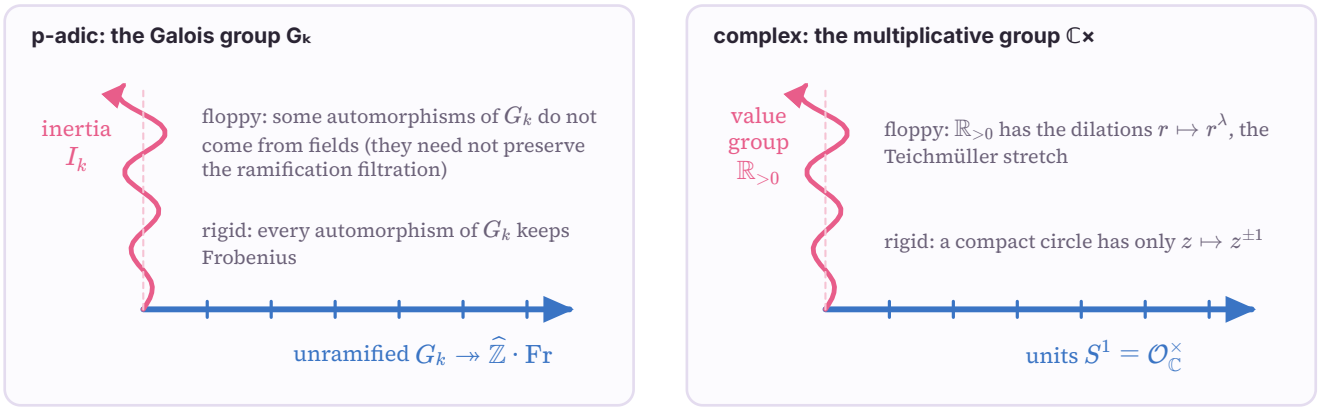


Figure 3. The two "combinatorial dimensions" of a local field [AbsTopIII, §I3, Rem 1.9.4]. G_k has cohomological dimension 2: the unramified direction is rigid (Frobenius is preserved by every automorphism), and the inertia direction is not. The complex numbers have the same split: units S^1 are rigid, the value group $\mathbb{R}_{>0}$ can be stretched. IUT's Θ-link will deliberately allow the stretch; that is the sense in which it is a "Teichmüller deformation".

So G_k alone is **mono-analytic**: it sees the multiplicative monoid and its two dimensions, but not how addition ties them together. IUT calls the ring structure the **arithmetic holomorphic structure**, by analogy with a complex structure on a real surface [Alien, §2.7 (vii)]. To get it back you must couple G_k to geometry, which is the job of Δ_X. Remark 1.9.4 of [AbsTopIII] says this exactly: being "coupled with Δ_X" via Π_X rigidifies both dimensions of G_k.

There is one more useful version of the input. If you know not just G_k but the pair $G_k \curvearrowright \mathcal{O}_{\bar{k}}^{\triangleright}$ (a group acting on an abstract monoid, called an MLF-Galois pair), then the Brauer-group argument above recovers the identification of the monoid's roots of unity with $\mu(G_k)$, and so $\mathrm{Aut}(G_k \curvearrowright \mathcal{O}^{\triangleright}) \cong \mathrm{Aut}(G_k)$. With only the units, $\mathrm{Aut}(G_k \curvearrowright \mathcal{O}^{\times}) \cong \widehat{\mathbb{Z}}^{\times} \times \mathrm{Aut}(G_k)$: a whole $\widehat{\mathbb{Z}}^{\times}$ of ambiguity appears [Alien, Ex 2.12.1–2.12.2]. Remember this $\widehat{\mathbb{Z}}^{\times}$; it returns in §11.

OTAKU NOTE: IDENTICAL TWINS

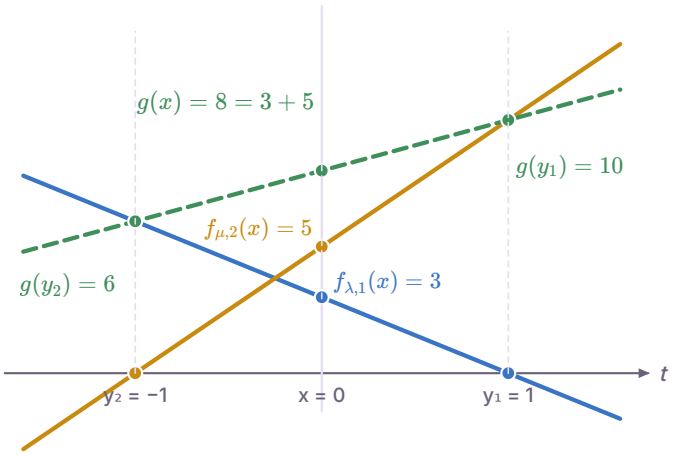
G_k is the classic manga twin problem. The twins have the same height, the same birthday, the same grades (p, d, q, f, e) and only a beauty mark tells them apart (the p-power roots of unity), and sometimes not even that. The detective's solution is always the same: look at how each twin interacts with the rest of the world. For a p-adic field, "the rest of the world" is a hyperbolic curve over it.

4 Algorithm B: Uchida's addition from multiplication 内田の補題

Suppose you have somehow recovered the *multiplicative* group $K_X^\times$ of the function field of a proper curve X over an algebraically closed field k, together with, for each point x, the valuation $\mathrm{ord}_x : K_X^\times \to \mathbb{Z}$ and the subgroup U_x of functions with value 1 at x. Then addition comes for free. This is "Uchida's lemma", implicit in Uchida's 1977 work on function fields and stated as an algorithm in [AbsTopIII, Prop 1.3].

ALGORITHM B · INPUT: $(K_X^\times,\ \{\mathrm{ORD}_X\},\ \{U_X\})$

1. Constants: $k^\times = \bigcap_x \ker(\mathrm{ord}_x)$. Evaluation: $\ker(\mathrm{ord}_x) = U_x \times k^\times$, so the projection to $k^\times$ is "the value at x".
2. The valuations let you talk about divisors, and so about $\Gamma^\times(D) = \{f : \mathrm{Div}(f) + D \geq 0\}$ and its dimension count $l(D)$, all multiplicatively [Prop 1.1].
3. Pick D with $l(D) = 2$ and three points x, y_1, y_2 as in [Prop 1.2]. For $\lambda \in k^\times$ let $f_{\lambda,1}$ be the element of $\Gamma^\times(D)$ with value λ at x and a zero at y_1; define $f_{\mu,2}$ likewise with a zero at y_2.
4. If $\lambda \neq -\mu$, the sum $f_{\lambda,1} + f_{\mu,2}$ is the *unique* $g \in \Gamma^\times(D)$ with $g(y_1) = f_{\mu,2}(y_1)$ and $g(y_2) = f_{\lambda,1}(y_2)$. Then $\lambda + \mu = g(x)$.
5. Evaluating at all points gives addition on all of $K_X^\times \cup \{0\}$.



THE TRICK, MULTIPLICATIVELY

$f_{\lambda,1} = 3(1 - t)$: value 3 at x, zero at y_1
$f_{\mu,2} = 5(1 + t)$: value 5 at x, zero at y_2
g: the *unique* element of $\Gamma^\times(D)$ with $g(y_1) = f_{\mu,2}(y_1)$ and $g(y_2) = f_{\lambda,1}(y_2)$
Then $g = f_{\lambda,1} + f_{\mu,2} = 8 + 2t$ and $\lambda + \mu = g(x)$. Finding g uses divisors, evaluation and $\times$ only.

Uchida-sensei's abacus: no + key needed.



Figure 4. Algorithm B on the simplest possible stage, $\mathbb{P}^1$ with $D = \infty$ (so $\Gamma(D) = \{a + bt\}$), $x = 0$, $y_1 = 1$, $y_2 = -1$, computing $\lambda + \mu$ for $\lambda = 3$, $\mu = 5$. The script finds $g = 8 + 2t$ from the two values $g(1) = 10$, $g(-1) = 6$ and reads off $g(0) = 8$. (Prop 1.2 is stated for curves of any genus; $\mathbb{P}^1$ is just the easiest place to draw it.)

Two lines on a graph do not look like deep mathematics, but notice what was *not* used: we never added two functions. We only used which functions vanish where, their values, and multiplication by constants. That is the whole philosophy of the subject in one picture. Mochizuki's summary of his reconstruction theorem is literally a sum of two terms [IUTchI, §I4]:

[AbsTopIII] §1 = Uchida's lemma + Belyi cuspidalization

and over finite fields the analogous slogan is "Uchida's lemma + the Lefschetz trace formula for Frobenius" [IUTchI, §I4]. The next three sections supply the inputs to Uchida's lemma from Π_X alone: the cyclotome (§5), the Kummer classes (§5) and the points (§6–7).

OTAKU NOTE: A SUDOKU THAT OUTPUTS ARITHMETIC

The puzzle magazine that gave sudoku its name has a rule that every grid must have a unique solution. Uchida's lemma is that rule applied to a linear system: two given values pin down one function, and its third value is the answer. You can do $3 + 5$ this way, and the build script does. It is the least efficient calculator in Japan, which is a competitive category.

5 Algorithm C: cyclotomes and Kummer classes 円分物とクンマー類

Synchronizing cyclotomes

Algorithm B needs functions, and to get functions out of a group you need Kummer theory, and Kummer theory needs coefficients: a copy of $\widehat{\mathbb{Z}}(1) = \varprojlim \mu_N$. A group gives you several candidates, called **cyclotomes**, and the first job is to identify them with each other canonically. For a proper curve X with a point x removed, [AbsTopIII, Prop 1.4] gives a group-theoretic isomorphism between

- the inertia group $I_x \cong \widehat{\mathbb{Z}}(1)$ of the puncture (the loop around the hole), and
- $M_X = \mathrm{Hom}(H^2(\Delta_X, \widehat{\mathbb{Z}}), \widehat{\mathbb{Z}})$, the dual of the "fundamental class" of the closed surface,

using the maximal cuspidally central extension $1 \to I_x \to \Delta_{X\setminus x}^{\text{c-cn}} \to \Delta_X \to 1$: its extension class is an element of $\mathrm{Hom}(M_X, I_x)$, and it is the isomorphism. Topologically: a small loop around the puncture is homologous to the boundary of the whole surface, so the "local" and "global" orientation are the same thing.

Kummer classes

Given a cyclotome, every invertible function f on an open $U \subseteq X$ has a Kummer class $\kappa_U(f) \in H^1(\Pi_U, M_X)$: the class of the torsor of all N-th roots of f, for all N. The base field must be **Kummer-faithful** (no nonzero element of any semi-abelian variety is infinitely divisible [AbsTopIII, Def 1.5]); every sub-p-adic field is [Rem 1.5.4]. Then [Prop 1.6] says:

1. κ_U is injective, so functions are faithfully recorded as cohomology classes;
2. restricting to the inertia groups I_x of the punctures gives an exact sequence $1 \to (k^\times)^\wedge \to H^1(\Pi_U, M_X) \to \bigoplus_{x\in S} \widehat{\mathbb{Z}}$, where the map to the sum reads off the order of zero or pole at each puncture;
3. the image of the functions is cut out by the condition that this divisor be principal, and principality is tested group-theoretically with the sections $G_k \to \Pi_J$ of the Jacobian.

Then [Prop 1.8] separates nonconstant functions from constants: a class is a nonconstant number-field function when some multiple of it vanishes at one point and not at another (values are computed by restricting to the decomposition group of the point, $\eta|_x = s_x^*(\eta) \in H^1(G_{k_x}, M_X)$), and a constant is a value of such a function at a point.

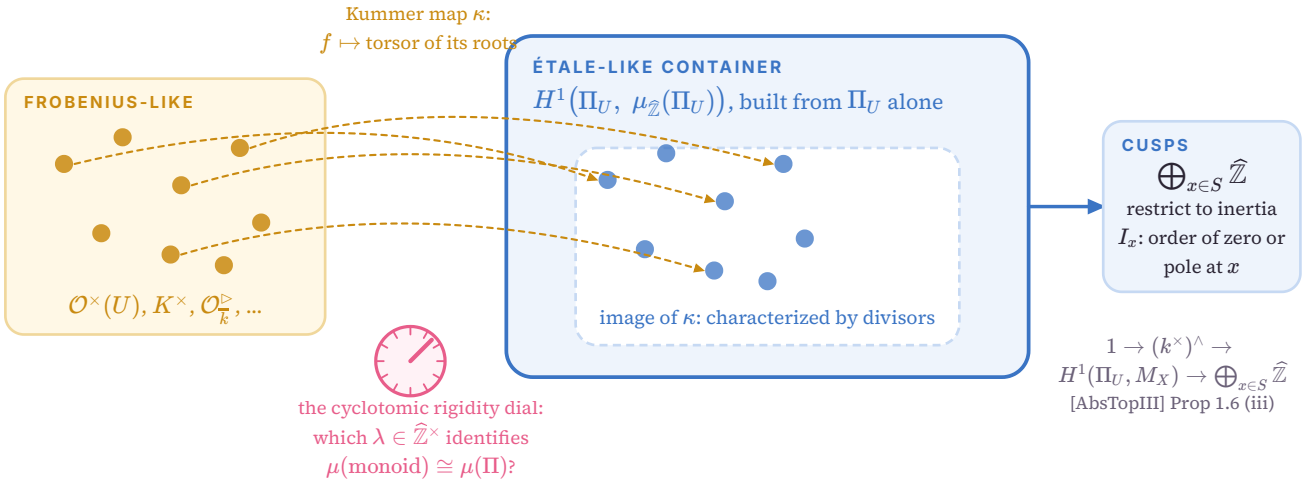


Figure 5. Kummer theory moves Frobenius-like data (functions, as elements of a monoid) into an étale-like container built from the group alone. The pink dial is the **cyclotomic rigidity** question: the torsion of the monoid and the cyclotome of the group are both copies of $\widehat{\mathbb{Z}}(1)$, but which isomorphism between them is the right one? An error by $\lambda \in \widehat{\mathbb{Z}}^\times$ would send f to "f^λ".

OTAKU NOTE: THE TUNING FORK AND THE TRANSFORMATION BELT

Before a school brass band plays, everyone tunes to one note. The cyclotome is that note: the inertia loop, the fundamental class of the surface and the roots of unity in the field all have to agree on pitch before anything else can be played. The Kummer map is the transformation belt of a long-running masked-rider franchise: insert a function, and it comes out as a cohomology class with the same personality and a much better costume. Cyclotomic rigidity is reading the belt's instructions so that you do not transform into f^{-1} by mistake.

6 Algorithm D: Belyi cuspidalization ベールイ・カスプ化

Uchida's lemma needs *points*: the valuations ord_x and the evaluations at x. In the group world a point x shows up as its decomposition group $D_x \subseteq \Pi_X$, but nothing in the group structure of Π_X singles that subgroup out (it is the image of a section of $\Pi_X \twoheadrightarrow G_k$, and there are many sections). The trick is to turn points into **cusps**, because cusps *are* visible: their inertia groups can be found group-theoretically [AbsTopI, Lemma 4.5; IUTchI, Rem 1.2.2]. "Cuspidalization" means constructing $\Pi_{X\setminus S} \twoheadrightarrow \Pi_X$ from Π_X alone. The tool is Belyi's theorem (1979): a curve over $\mathbb{C}$ is defined over a number field exactly when it admits a map to $\mathbb{P}^1$ branched over at most three points.

The idea [AbsTopII, Ex 3.6, Cor 3.7]: take the point x you want and a Belyi map $\beta\colon W \to P$, where $P = \mathbb{P}^1 \setminus \{0, 1, \infty\}$, $W \subseteq U \subseteq V$ are open, and x is one of the points β sends to $\{0, 1, \infty\}$. Over P the map is finite étale, so Π_W is an *open subgroup* of Π_P, which you can find by group theory. Then $W \subseteq U \setminus \{x\}$ is an open immersion, so $\Pi_W \twoheadrightarrow \Pi_{U\setminus\{x\}}$ is a quotient by inertia groups of cusps, which you can also find. A chain of "cover, then fill in cusps" operations takes you from Π_P to $\Pi_{U\setminus\{x\}}$.

ALGORITHM D · [ABSTOPII] COR 3.7 · INPUT: $1 \to \Delta \to \Pi \to G \to 1$, X OF STRICTLY BELYI TYPE

1. (a) Find a normal open $\Pi_V \subseteq \Pi$ and a chain of type $\subset, \subset, \bullet, \ldots, \bullet, \subset, \bullet, \ldots, \bullet, \supset$ (covers $\subset$, de-cuspidalizations $\bullet$) ending in a terminal isomorphism with the trivial chain. It yields $\Pi_U \twoheadrightarrow \Pi_V$ for $U = V \times_X U_X$.
2. (b) Descend to $\Pi_{U_X} \twoheadrightarrow \Pi$ using the unique lift of the outer action of Π/Π_V.
3. (c) The decomposition groups of the points of $X \setminus U_X$ are the images of the cuspidal decomposition groups of Π_{U_X}.

"Strictly Belyi type" means defined over a number field and isogenous to a hyperbolic curve of genus 0 [AbsTopII, Def 3.5]; every finite étale cover of a once-punctured elliptic curve over a number field qualifies [Alien, Ex 2.12.3 (i)]. The method reaches points defined over number fields; over a p-adic field all closed points follow by p-adic approximation [AbsTopII, Rem 3.7.2].

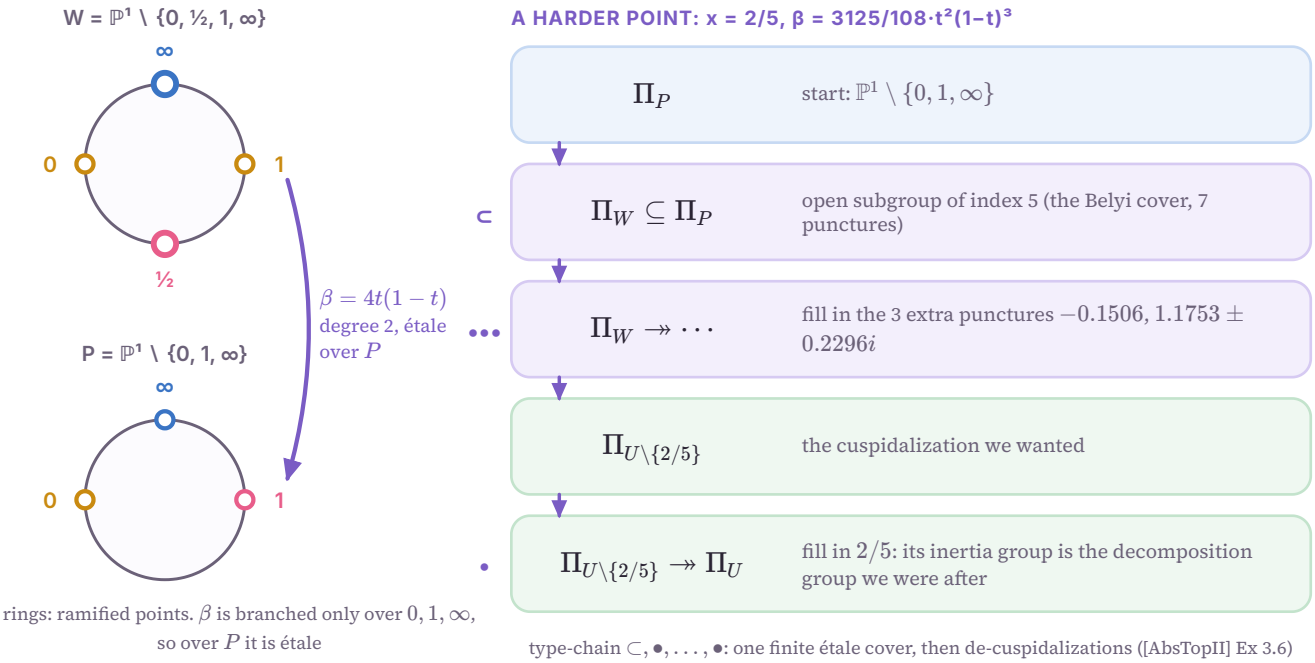


Figure 6. Left: $\beta(t) = 4t(1-t)$ sends $0, 1 \mapsto 0$, $\frac{1}{2} \mapsto 1$ and $\infty \mapsto \infty$, and its only critical points are $\frac{1}{2}$ and ∞. So $\mathbb{P}^1 \setminus \{0, \frac{1}{2}, 1, \infty\} \to \mathbb{P}^1 \setminus \{0, 1, \infty\}$ is finite étale of degree 2, and π_1 of the tripod minus $\frac{1}{2}$ is an index-2 subgroup of π_1 of the tripod. (By Schreier's formula an index-d subgroup of a free group of rank r is free of rank $d(r-1)+1$; here $2 \cdot 1 + 1 = 3$, matching four punctures.) Right: the point $\frac{2}{5}$ needs Belyi's polynomial $\beta = 3125/108 \cdot t^2(1-t)^3$ of degree 5, which also sends three other points to 1; they are filled back in by de-cuspidalization.

Belyi's own algorithm makes every algebraic point reachable: first push the point to a rational one by repeatedly composing with minimal polynomials, then use the map $t \mapsto \frac{(m+n)^{m+n}}{m^m n^n} t^m (1-t)^n$, which is ramified only over $0, 1, \infty$ and sends $\frac{m}{m+n} \mapsto 1$. For $\frac{2}{5}$ the script checks that $\beta(\frac{2}{5}) = 1$, that $\frac{2}{5}$ is a double root of $\beta = 1$, that the other three roots are -0.1506 and $1.1753 \pm 0.2296\,i$, and that Schreier's formula gives $5 \cdot 1 + 1 = 7 - 1$, as it must for 7 punctures. Another example: $4(t-1)/t^2$ does the same job for the point 2.

OTAKU NOTE: ORIGAMI AND THE ROLLING BALL

A Belyi map is origami for Riemann surfaces: you fold the whole sphere onto itself until the point you care about lands on one of three corners, and the creases (the branch points) all sit on the corners too. There is also a well-loved PlayStation game in which a tiny prince rolls a sticky ball over everything until the whole world is stuck to it. Belyi's algorithm is that ball: point after point gets rolled into $\{0, 1, \infty\}$, and none of them ever falls off. The King of All Cosmos would approve of the degree.

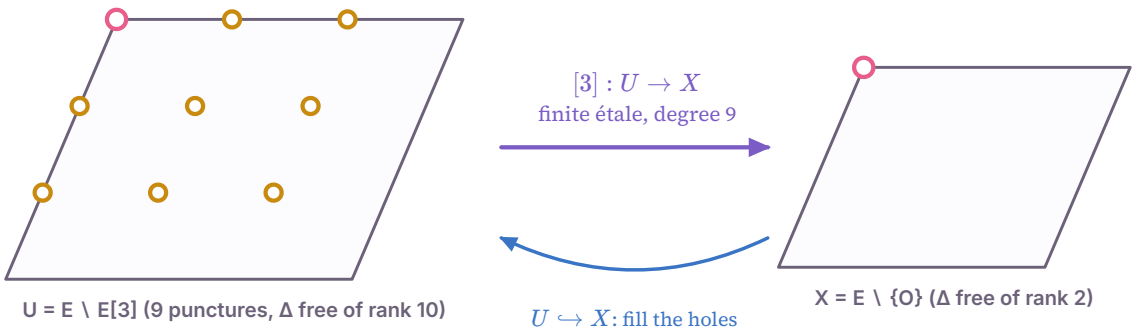
7 Algorithm E: elliptic cuspidalization 楕円カスプ化

For a once-punctured elliptic curve $X = E \setminus \{O\}$ there is a second, more elementary way to turn points into cusps, which works for torsion points and also in pro-Σ settings where Belyi maps are unavailable [AbsTopII, Ex 3.2, Cor 3.3]. Multiplication by N on E restricts to a finite étale map

$$[N]\colon\ U = E \setminus E[N] \ \longrightarrow\ E \setminus \{O\} = X \qquad \text{of degree } N^2.$$

So Π_U is an open subgroup of Π_X of index N^2, visible to group theory. But U is also an open subscheme of X, so $\Pi_U \twoheadrightarrow \Pi_X$ by filling in the $N^2 - 1$ nonzero torsion points. The same group plays both roles; following the chain "cover by $[N]$, then de-cuspidalize $N^2 - 1$ times" recovers the cuspidalization $\Pi_U \twoheadrightarrow \Pi_X$ and with it the decomposition groups of the N-torsion points. The general statement handles

any "elliptically admissible" orbicurve: one whose k-core is semi-elliptic, i.e. a quotient of a once-punctured elliptic curve by ± 1 [AbsTopII, Def 3.1].



THE SAME GROUP, TWO JOBS

Π_U is an open subgroup of Π_X of index 9 (via $[3]$) *and* it surjects onto Π_X (via $U \subseteq X$). So Π_X alone produces the cuspidalization $\Pi_U \twoheadrightarrow \Pi_X$ and the decomposition groups of the torsion points. For 11a1 with $N = 2$ the four removed points are O and the points with $x = 4.3463, -1.6732 \pm 1.3208i$.

Figure 7. Elliptic cuspidalization for $N = 3$. Schreier's formula checks the ranks: an index-N^2 subgroup of a free group of rank 2 has rank $N^2 + 1$, which is 5 for $N = 2$ and 10 for $N = 3$, the rank of π_1 of a torus with N^2 punctures. The 2-torsion of 11a1 ($y^2 + y = x^3 - x^2 - 10x - 20$) is computed by the build script from $4x^3 - 4x^2 - 40x - 79 = 0$.

In IUT this is how the **evaluation points** are reached: the theta function is evaluated at μ_--translates of cusps, which are torsion points, and their decomposition groups are reconstructed by elliptic cuspidalization [Alien, §3.4 (iii)]. It is also behind [IUTchI, Cor 1.2], which reconstructs $\Pi_{\underline{X}}$, $\Pi_{\vec{C}}$, $\Pi_{\underline{C}}$ and the cusps $0, \epsilon', \epsilon''$ from $\Pi_{\vec{X}}$, and recovers the prime l itself from an index count.

THE TOOLKIT SAYS · IUT RUN RECONSTRUCTCOVERINGS --ABC 1 2400 2401 --L 13

```
step 2: reconstruct Π_C ⊇ Π_X→ and Δ_X ⊆ Δ_C ⊆ Π_C ([AbsTopII] Cor 3.3 (i),(ii))
step 3: recover l from l² = [Δ_X : Δ_X_]·[Δ_X_ : Δ_X→] = [Δ_X : Δ_X→] = [Δ_C : Δ_X→]/2
        l² = 169 = 13·13 = 338/2
```

The process walks the eight steps of the proof of [IUTchI] Cor 1.2 on the example $1 + 2400 = 2401, l = 13$, and checks the index identity. It records the steps; it does not manipulate profinite groups.

OTAKU NOTE: SHADOW CLONES

$[N]^{-1}(O) = E[N]$: one origin becomes N^2 of them, every one an exact copy. Any ninja anime fan knows the technique. The point of the jutsu here is the opposite of the usual one: instead of confusing the enemy, it lets the group theorist find every clone, because each clone is a hole you can put your finger through.

8 The full algorithm: [AbsTopIII] Theorem 1.9 復元定理

Now everything plugs together. Let X be a hyperbolic orbicurve of strictly Belyi type over a sub-p-adic field k (a number field or a p-adic field, say). [AbsTopIII, Thm 1.9] gives a functorial group-theoretic algorithm that reconstructs, from the extension $1 \to \Delta_X \to \Pi_X \to G_k \to 1$ alone, the "NF-portion of the function field": the field $\overline{k}_{\mathrm{NF}}$ of algebraic numbers in $\overline{k}$ and the function fields of the covers of X over it, *with their field structures*. When k is a number field or an MLF the extension can be replaced by the single group Π_X [Rem 1.9.2; AbsTopI, Thm 2.6 (v), (vi)].

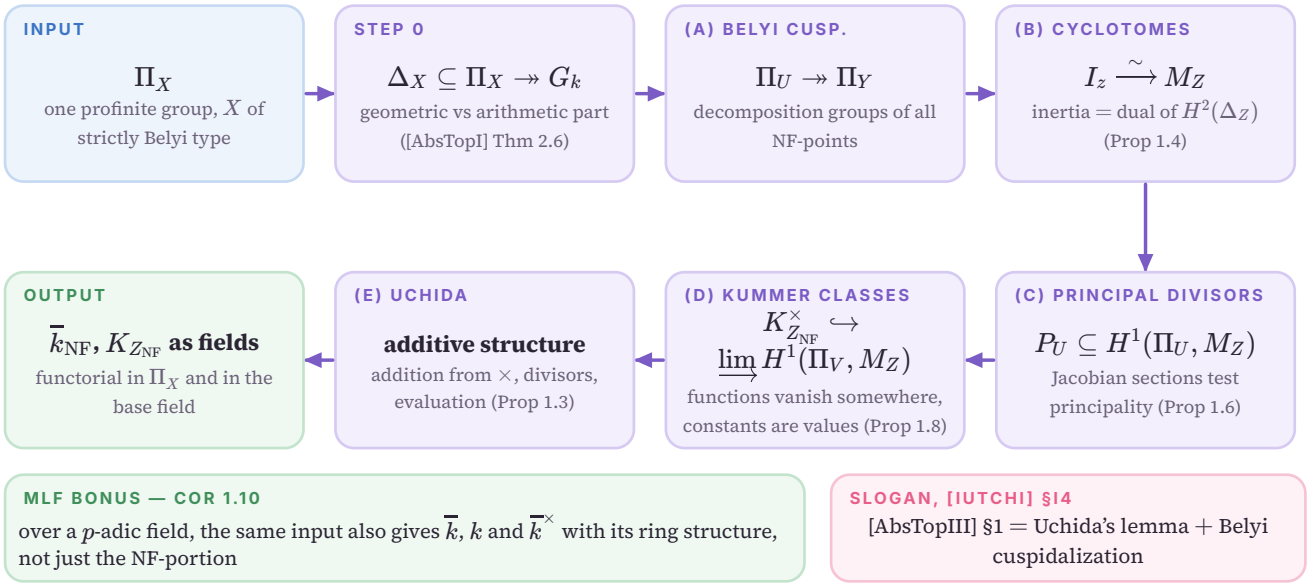


Figure 8. The steps (a)–(e) of [AbsTopIII, Thm 1.9]. Steps (a)–(d) make the inputs of Uchida's lemma out of the group; step (e) is Uchida's lemma. The output is functorial with respect to open injections of groups and to change of base field, including from a number field to its completion; that last property is what lets IUT apply the same algorithm prime by prime [Rem 1.9.5 (ii)].

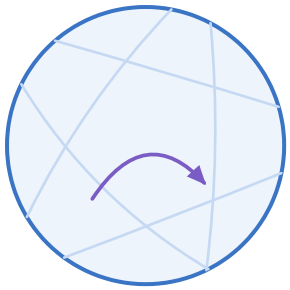
Three remarks from the paper are worth keeping in your pocket.

- **Combinatorialization** [Rem 1.9.7]: the algorithm is "a complete combinatorialization, independent of the base field, of the orbicurve", in the same spirit as replacing a scheme's covers by an abstract Galois category, or a log-regular scheme's local rings by monoids.
- **An elementary version** [Thm 1.11]: if you drop Belyi cuspidalization and only want the function field, there is a much simpler algorithm valid over any Kummer-faithful field. Mochizuki hopes it can be taught to advanced undergraduates.
- **MLF bonus** [Cor 1.10]: over a p-adic field, Π_X also gives back $\overline{k}$, k and the ring structure of $\overline{k}$, not only its algebraic-number part. This is the local result IUT uses at each prime.

OTAKU NOTE: THE CULINARY BATTLE
In a famous cooking show the chairman unveils a single ingredient and the chefs must build a whole menu from it in an hour. Theorem 1.9 is that show with Π_X under the cloche. Course one, the Galois group (Algorithm A); course two, a cyclotome reduction (§5); the main course, Belyi-folded points with a Kummer glaze (§6); dessert, Uchida's addition. The judges taste it in another universe and it is the same dish. Allez cuisine!

9 The archimedean side: Aut-holomorphic spaces アルキメデス的側面

A number field also has complex places, and there is no profinite Galois group there. [AbsTopIII, §2] supplies the archimedean counterpart: an **Aut-holomorphic space** is a topological surface together with, for every connected open U, the subgroup $\mathrm{Aut}^{\mathrm{hol}}(U) \subseteq \mathrm{Homeo}(U)$ of holomorphic automorphisms [Def 2.1]. The key fact is that the holomorphic-or-antiholomorphic automorphisms of the unit disc form a commensurably terminal subgroup of its homeomorphisms [Prop 2.2], so this weak-looking data pins down the complex structure up to complex conjugation. A refinement of the earlier "parallelograms, rectangles, squares" method then recovers the local linear holomorphic structure [Prop 2.5, 2.6], and elliptic cuspidalization on the Riemann surface ties it to the Galois-theoretic theory of §1 [Cor 2.7–2.9].



$\mathrm{Aut}^{\pm\mathrm{hol}}(\mathbb{D}) \subseteq \mathrm{Homeo}(\mathbb{D})$ is commensurably terminal

"parallelograms, rectangles, squares": the linear and then holomorphic structure is read off from which shapes the topology can see

ARCHIMEDEAN DICTIONARY

profinite group ↔ **Aut-holomorphic space**: a surface plus, for each open U, the subgroup of holomorphic automorphisms

$\widehat{\mathbb{Z}} \cdot \mathrm{Fr}$ rigid ↔ S^1 rigid

I_k floppy ↔ $\mathbb{R}_{>0}$ floppy

Belyi / elliptic cuspidalization ↔ elliptic cuspidalization on the Riemann surface ([AbsTopIII] Cor 2.7–2.9)

Figure 9. The archimedean dictionary. The unit circle $S^1 = \mathcal{O}_{\mathbb{C}}^{\times}$ plays the part of the rigid Frobenius direction and $\mathbb{R}_{>0}$ the floppy inertia direction (Figure 3).

OTAKU NOTE: THE GHOST AND THE SHELL

In a famous cyberpunk manga, a cyborg's "ghost" can move between bodies, and the question is what survives the move. Here the shell is the topological surface, which anyone can copy, and the ghost is the holomorphic structure. The Aut-holomorphic structure is a way of carrying the ghost in a form that the shell can hold: not the complex coordinate itself, but the list of motions that respect it.

10 Étale-like, Frobenius-like, and mono-anabelian transport エタール的とフロベニウス的

The prototype in positive characteristic

Mochizuki calls the following example "the fundamental model, or prototype, for a quite substantial portion of inter-universal Teichmüller theory" [Alien, Ex 2.6.1]. Let K be the function field of a curve over a finite field of characteristic p. The Frobenius $\Phi\colon f \mapsto f^p$ has degree p, so it is far from an isomorphism, yet it induces an *isomorphism* $\Psi\colon G_K \xrightarrow{\sim} G_K$ of Galois groups, because it is purely inseparable. There are two ways to rebuild K from G_K: by class field theory (which uses the ring structure heavily) and by Kummer theory (which uses only the multiplicative monoid and one cyclotomic rigidity isomorphism λ). Either way, the square below cannot commute.

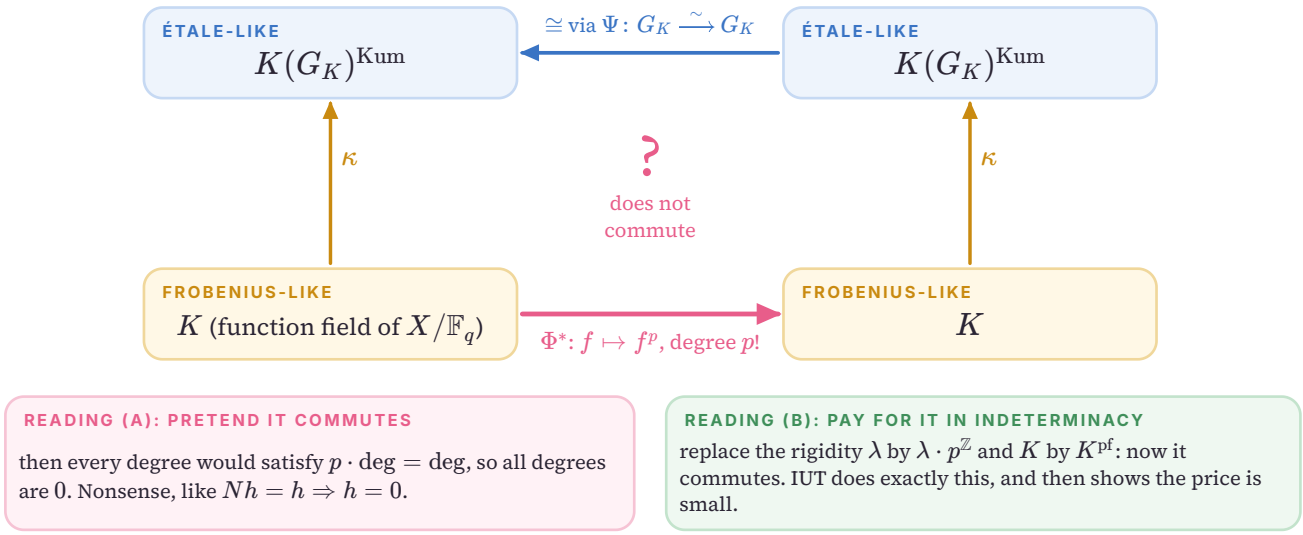


Figure 10. [Alien, Ex 2.6.1]. The vertical maps and the top map are isomorphisms; the bottom map multiplies degrees by p. Reading (b) is IUT's strategy in miniature: admit just enough indeterminacy to make the diagram commute, then show the indeterminacy is small, and conclude $N \cdot h \leq h + C$, so $h \leq C/(N-1)$.

The vocabulary

From this example IUT takes its terms [Alien, §2.7]. A **link** is a wall between two "mutually alien" copies of scheme theory (the Θ-link and the log-link). **Frobenius-like** objects are the ones used to set up the link; they live on one side only, carry "positive mass" (degrees, heights) and are changed by the link, like K under Φ. **Étale-like** objects are groups such as G_K, Π_X, and everything built from them by an algorithm; they are "zero mass" *containers*, mapped isomorphically across the link, though only by an unknown isomorphism. An object that maps isomorphically across a link is **coric**.

Mono-anabelian transport computes what the link does to positive-mass data in three moves: (1) Kummer theory passes from Frobenius-like to étale-like objects (*Kummer-detachment*); (2) the anabelian algorithm, being the same on both sides, carries the étale-like objects across (*étale-transport*); (3) inverse Kummer theory re-attaches Frobenius-like objects on the far side. Each move has its own indeterminacy: Kummer-detachment indeterminacies (mostly about cyclotomic rigidity) and étale-transport indeterminacies (the étale-like objects are only known up to isomorphism) [IUTchIII, Rem 1.5.4].

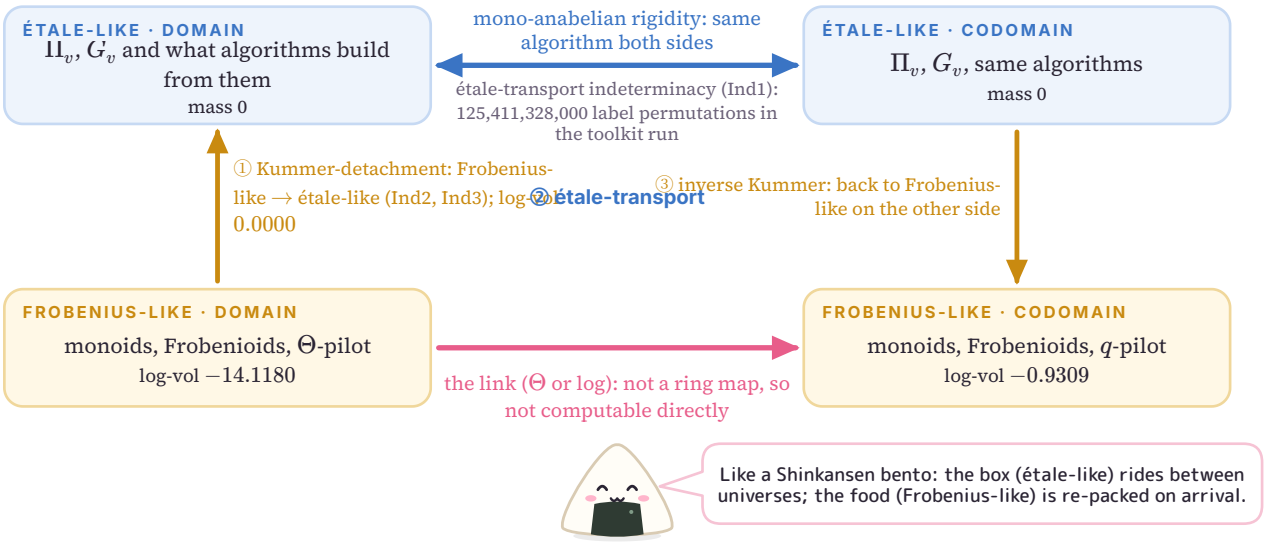


Figure 11. [Alien, Fig 2.1] with the toolkit's run on $1 + 2400 = 2401$, $l = 13$: the Θ-pilot log-volume -14.1180 goes to mass 0 when detached, travels as étale-like data, and the codomain is re-attached with the q-pilot log-volume -0.9309.

THE TOOLKIT SAYS · IUT RUN TRANSPORTMONOANABELIAN --ABC 1 2400 2401 --L 13

```
run: Frobenius-like input at (0,0): Θ-pilot log-volume                -14.1180
run: Kummer-detach at (0,0): pass to étale-like (0,∘) data (mass 0)    0.0000
run: étale-transport across the Θ-link via D⊢ (Ind1 permutations)      125411328000
run: re-attach at (1,0): q-pilot log-volume (the mass of the codomain) -0.9309
```

The third number is a count (the size of the label-permutation indeterminacy the toolkit tracks), not a log-volume.

The same example also explains the two words **arithmetic holomorphic** and **mono-analytic** [Alien, §2.7 (vii)]. The class-field-theory reconstruction depends on the whole ring; the Kummer reconstruction depends only on the multiplicative monoid plus λ. Kummer theory "isolates the dependence on the ring structure in the compact form of the cyclotomic rigidity isomorphism". That is why the next section is entirely about λ.

OTAKU NOTE: EKIBEN ON THE BULLET TRAIN

A station bento (*ekiben*) is a box designed to survive a high-speed journey. The box is étale-like: rigid, standard, the same in Tokyo and in Hakata, weighing nothing that matters. The food is Frobenius-like: it is what you actually care about, and it is different in every region. Mono-anabelian transport ships the boxes and re-fills them on arrival, and the indeterminacies are the passengers' complaints that the karaage moved around in transit. They are allowed to complain only up to a bounded amount.

11 Cyclotomic rigidity, three ways 円分剛性の三つの方法

Everything Frobenius-like gets into the étale-like world through Kummer theory, and Kummer theory is only as good as its cyclotomic rigidity isomorphism: the identification of the roots of unity of the monoid with the cyclotome of the group. IUT needs this identification to be **multiradial**: it must survive arbitrary automorphisms of the coric data on the other side of the Θ-link. That data contains $G_v \curvearrowright \mathcal{O}^{\times\mu}$, and $\widehat{\mathbb{Z}}^\times$ acts on it (raise units to the power λ); this action moves the roots of unity of the monoid but not the cyclotome of the group. So any rigidity that depends on the unit group is ruined by the Θ-link, and IUT must *decouple* the value-group data from the units [Alien, §3.4 (i)]. It uses three methods, summarized in [Alien, Fig 3.10]:

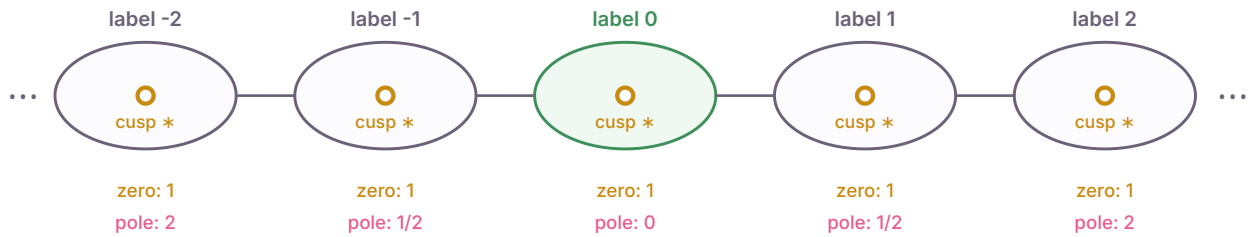
method	applied to	how it fixes λ	radiality	profinite topology
Brauer groups / local class field theory for $G_k \curvearrowright \mathcal{O}_k^{\triangleright}$	local units (a_Θ)	$\mathbb{Q}/\mathbb{Z} \cong H^2(G_k, \mu)$ built from the monoid and from the group must agree [Alien, Ex 2.12.1 (ii)]	uniradial	compatible
mono-theta environments	local value groups at bad v (b_Θ)	commutators in the theta group of the line bundle on the Tate curve; zeros of order exactly 1 [EtTh, Cor 2.19]	multiradial	compatible
κ-coric rational functions	global F_{mod} (c_Θ)	$\mathbb{Q}_{>0} \cap \widehat{\mathbb{Z}}^\times = \{1\}$: only $\pm\lambda$ keeps Kummer classes of functions inside the reconstructed function field [IUTchI, Ex 5.1 (v)]	multiradial	incompatible

The global method: κ-coric functions

The global realified Frobenioids that cross the Θ-link are, in essence, arithmetic line bundles on copies of the number field F_{mod}. Elements of F_{mod} are produced as *values of functions at special points*, exactly as in [AbsTopIII, §1] [Alien, §3.4 (ii)]. The functions used are the **κ-coric rational functions** on the coarse curve $|C_{F_{\mathrm{mod}}}| \cong \mathbb{A}^1$: those that take root-of-unity values at the "strictly critical" points (the nonzero 2-torsion points) [IUTchI, Rem 3.1.7]. That condition can be checked on the étale-like side by restricting Kummer classes to decomposition groups, and it is what decouples them from the local units. Their rigidity comes from the elementary fact $\mathbb{Q}_{>0} \cap \widehat{\mathbb{Z}}^\times = \{1\}$: if $\lambda \in \widehat{\mathbb{Z}}^\times$ sends the Kummer class of every function to the class of a function, then looking at orders of zeros (positive integers) forces $\lambda = \pm 1$ [Alien, Ex 2.13.1 (ii)]. The leftover ± 1 is harmless because $F_{\mathrm{mod}}^\times$ is closed under inversion. The price is that the argument only works for the full profinite group, not for its finite quotients; that does not matter globally [Alien, §3.4 (v)].

The local method: mono-theta environments

At a bad prime v, X_v is a once-punctured Tate curve and its tempered cover Y_v has special fibre an infinite chain of projective lines (Figure 12). The theta function $\ddot{\Theta}_v$ has a simple zero at every cusp, which allows a "mod N" version of the $\mathbb{Q}_{>0} \cap \widehat{\mathbb{Z}}^\times$ argument with no ± 1 and compatible with the topology. But its poles are not symmetric under the $\mathbb{Z}$-action, so the naive version fails **discrete rigidity**. The fix is to package the data as a triple (the $\mathbb{G}_m$-torsor of the line bundle, its automorphisms by symmetries and constants, the theta section): the **mono-theta environment** [EtTh, Def 2.13]. It is invariant under the $l \cdot \mathbb{Z}$-symmetries, and commutators of its automorphisms recover cyclotomic rigidity; all of this is "essentially a formal consequence of the quadratic structure of the commutators of the theta groups" [Alien, §3.4 (iv)]. Mono-theta environments have étale-like versions (from Π_v^{tp}) and Frobenius-like versions (from a tempered Frobenioid), which is what lets the theta values travel by mono-anabelian transport [IUTchII, Prop 1.2, Cor 1.10–1.12].



special fibre of the tempered cover $Y_v \to X_v$ (Galois group $\cong \mathbb{Z}$): a chain of $\mathbb{P}^1$s glued ∞ to 0. The theta function has a simple zero at every cusp and a pole of order $j^2/2$ (in units of $\mathrm{ord}(q_v)$) on component j. The zeros are symmetric under $\mathbb{Z}$; the poles are not. That asymmetry is why one packages the data as a mono-theta environment.

Figure 12. After [Alien, Fig 3.9]. The cusps (orange) are symmetric; the pole orders $j^2/2$ (pink) are a parabola, and a parabola is not translation invariant. The quadratic exponent is the same one that makes Gaussian integrals work, which is the theme of [Alien].

OTAKU NOTE: THREE KEYS TO ONE DOOR

A classic adventure series on a certain handheld always hides the last dungeon behind a door with three locks. The first key (class field theory) opens it, but it only works from inside the room, which is useless if you are in another universe. The second key (mono-theta) works from anywhere and fits the lock exactly, but only at the bad primes. The third key (κ-coric functions) works globally, but it only turns if you hold the entire profinite group at once, never a finite piece of it. IUT carries all three, and its inventory screen is Fig 3.10.

12 The log-link and log-shells 対数リンクと対数殻

The p-adic logarithm $\log_k : \mathcal{O}_k^\times \to k$ is not a ring map (it turns $\times$ into $+$), but it is G_k-equivariant. That makes it a link in the sense of §10, with Π_X coric [AbsTopIII, §I1; Alien, Ex 2.12.3]: in the chain $\cdots \xrightarrow{\log} \mathcal{O}_k \xrightarrow{\log} \mathcal{O}_k \xrightarrow{\log} \cdots$, each copy of $\mathcal{O}_k$ belongs to a different universe, but the same Π_X sits over all of them, and Corollary 1.10 rebuilds the same field structure from each. The diagram of Kummer maps and logs does not commute as maps of elements, but it does commute "as regions". Put

$$\mathcal{I} = (2p)^{-1} \cdot \log_k(\mathcal{O}_k^\times) \subseteq k, \qquad \mathcal{O}_k \subseteq \mathcal{I} \supseteq \log_k(\mathcal{O}_k^\times).$$

Every composite of arrows, through any number of log-iterates, lands inside the étale-like log-shell $\mathcal{I}(\Pi_X)$. This is **upper semi-commutativity**, and together with the fact that log-volumes are compatible with log it is all IUT needs from the log-link [Alien, Ex 2.12.3 (iv); IUTchIII, Prop 1.2, Rem 1.2.2]. The log-shell comes in four flavours: holomorphic or mono-analytic, Frobenius-like or étale-like; the mono-analytic ones are built from $\mathcal{O}^{\times\mu}$ alone, since $\mathcal{O}^{\times\mu} \otimes \mathbb{Q} \cong \mathcal{I} \otimes \mathbb{Q}$.

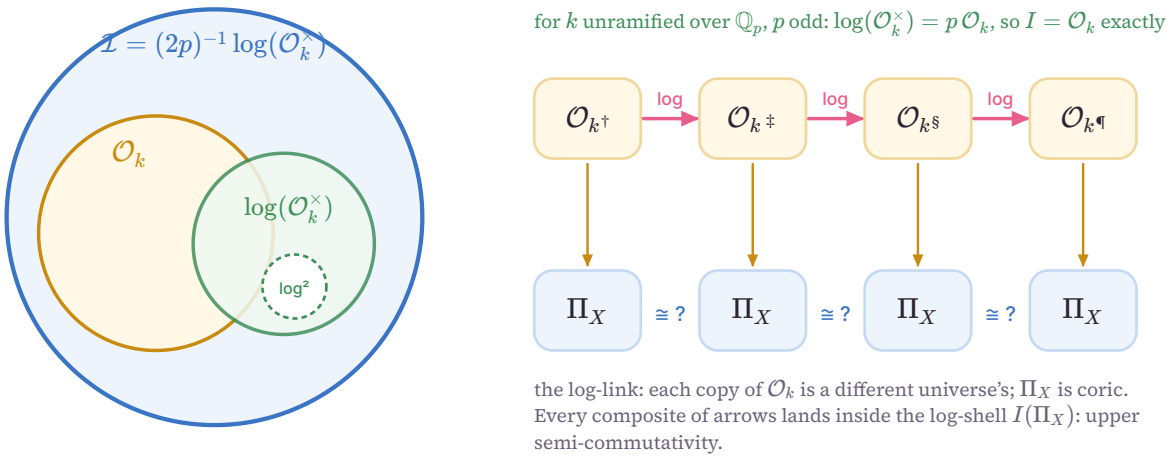


Figure 13. Log-shells. For k unramified over $\mathbb{Q}_p$ with p odd, log maps $1 + p\mathcal{O}_k$ isomorphically onto $p\mathcal{O}_k$ and kills the roots of unity, so $\mathcal{I} = (2p)^{-1}p\,\mathcal{O}_k = \mathcal{O}_k$: log-volume 0. Ramification makes the shell strictly bigger than $\mathcal{O}_k$.

OTAKU NOTE: THE FUTON AND THE TIME LOOP

The log-link is a time-loop anime: every episode the world resets, and the protagonist keeps only what fits in a small container. The log-shell is that container, and its whole job is to be big enough that nothing from any previous loop sticks out. For an unramified field and an odd prime it fits exactly, like a futon folded into its closet. Ramification is when the futon is a little too big and the closet door will not quite shut, which is what the $\log(\mathfrak{d})$ terms in [IUTchIV] keep track of.

13 Where IUT calls each algorithm IUTの中での呼び出し

Here is the call graph. Each Hodge theater carries arithmetically holomorphic data, and each reconstruction algorithm is applied to it *inside* that theater; the Θ-link lets only mono-analytic data through, so on the far side only algorithms that start from G_v (Algorithm A, mono-analytic log-shells) can be run on the shared data. The vertical log-links share the holomorphic Π_v, so the full algorithms can be run along a whole column of the log-theta-lattice.

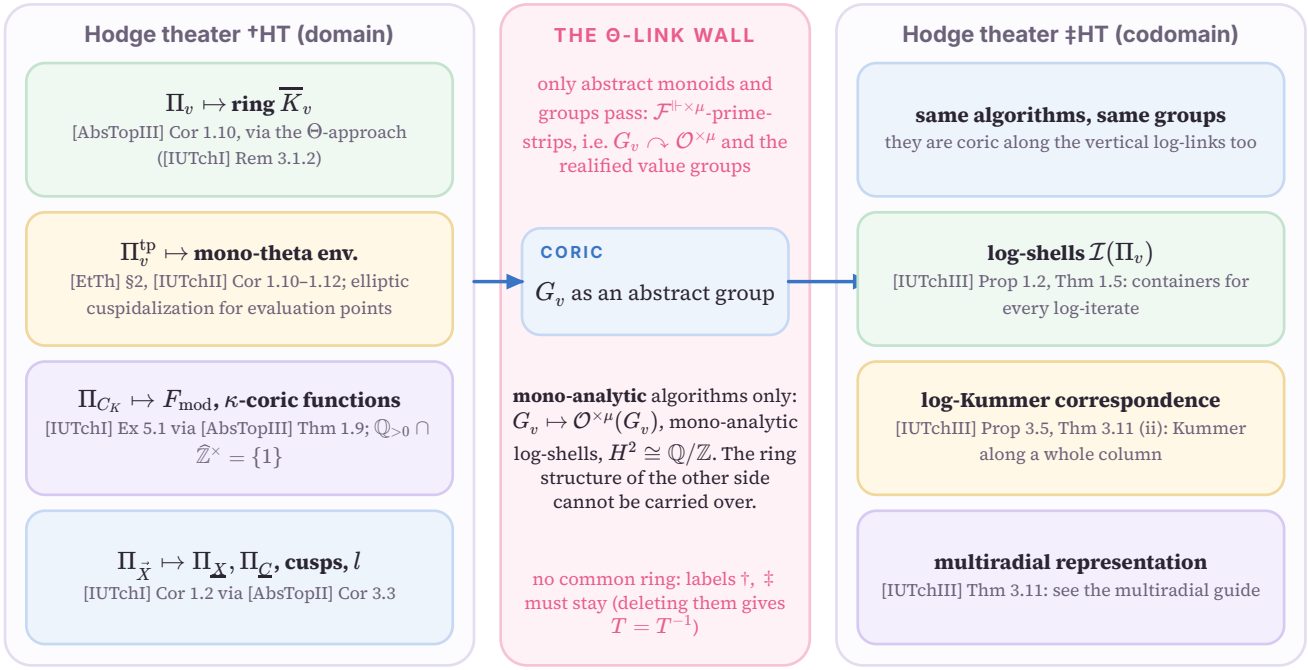


Figure 14. Where the algorithms are called. Left: inside the domain Hodge theater. Middle: what crosses the Θ-link. Right: what the codomain does with it, leading to the multiradial representation of [IUTchIII, Thm 3.11], which is the subject of the companion guide `multiradial_guide.pdf`.

where	input group	algorithm	output used for
[IUTchI] Cor 1.2	$\Pi_{\vec{X}}$ at each v	elliptic cuspidalization [AbsTopII, Cor 3.3]	$\Pi_{\underline{X}}, \Pi_{\underline{C}}$, cusps $0, \epsilon', \epsilon''$, the prime l
[IUTchI] Rem 3.1.2	$\Pi_{\underline{X}_K} \subseteq \Pi_{\underline{C}_K}$	Thm 1.9 applied "via the Θ-approach"	the cyclotome $M_X \cong l \cdot \Delta_\Theta$ of the punctured elliptic curve
[IUTchI] Ex 5.1	$\pi_1(^{\dagger}\mathcal{D}^{\circledast})$	Thm 1.9, Belyi cuspidalization, κ-coric functions	$\overline{F}$, F_{mod}, the global Frobenioids
[IUTchI] Def 5.2 (v)–(vi)	$\pi_1(^{\ddagger}\mathcal{D}_v)$	Cor 1.10 (local)	$\mathcal{O}^{\triangleright}_{F_v}$, local κ-coric pseudo-monoids
[IUTchII] §1	Π_v^{tp}, v bad	mono-theta environments [EtTh]	cyclotomic, discrete and constant multiple rigidity
[IUTchII] §2–3	Π_v^{tp}	elliptic cuspidalization at evaluation points	Galois evaluation of theta values $q_v^{j^2}$
[IUTchIII] Prop 1.2, Thm 1.5	G_v, Π_v	Algorithm A, Cor 1.10, log	the four kinds of log-shells; bi-coric data
[IUTchIII] Thm 3.11	all of the above	mono-anabelian transport along the log-theta-lattice	the multiradial representation of the Θ-pilot

14 Cheat sheet, honest notes, references まとめ

algorithm	input	output	key trick	analogy
A · local invariants	G_k	p, d, q, f, e, I_k, Frobenius, cyclotome, $H^2 \cong \mathbb{Q}/\mathbb{Z}$	local class field theory on G^{ab}	the twins' report cards
B · Uchida	$K^\times$, valuations, evaluations	addition	a two-dimensional linear system	the sudoku calculator
C · cyclotomes, Kummer	Π_U, Δ_X	$I_x \cong M_X$; functions as classes	the cuspidally central extension; principal divisors via the Jacobian	tuning fork, transformation belt
D · Belyi cusp.	Π_X, strictly Belyi type	$\Pi_{X\setminus S} \twoheadrightarrow \Pi_X$, points as decomposition groups	Belyi maps are étale over the tripod	origami; the sticky ball
E · elliptic cusp.	Π_X, elliptically admissible	the same, for torsion points	$[N]$ is étale off $E[N]$	shadow clones
Thm 1.9 / Cor 1.10	Π_X	$\overline{k}_{\mathrm{NF}}$ (resp. $\overline{k}$) and function fields as fields	A + C + D, then B	the culinary battle
Aut-holomorphic	a surface and its local automorphism groups	the complex structure, up to conjugation	disc automorphisms are commensurably terminal	ghost and shell
transport	Frobenius-like data on one side	its description on the other side, with indeterminacies	Kummer, anabelian rigidity, inverse Kummer	ekiben
cyclotomic rigidity	monoid with group action	$\mu(\mathrm{monoid}) \cong \mu(\mathrm{group})$	Brauer groups; theta commutators; $\mathbb{Q}_{>0} \cap \widehat{\mathbb{Z}}^\times = \{1\}$	three keys
log-shells	$\mathcal{O}_k^\times$, Π_X	a compact container for all log-iterates	upper semi-commutativity	the futon

HONEST NOTES

What the toolkit does. The C toolkit and the web app model IUT's data types and processes and keep their bookkeeping (labels, indices, log-volumes, indeterminacy counts). They do not compute with profinite groups, and no finite program can run Theorem 1.9 on an actual Π_X. The numbers in this guide come either from that bookkeeping or from the small worked examples in `build.js`, which are checked when the PDF is built.

What is and is not contested. The anabelian papers surveyed here ([AbsAnab], [AbsTopI–III], [EtTh]) were published before and independently of IUT. The disagreement that followed the March 2018 discussions between Scholze, Stix and Mochizuki concerns the proof of [IUTchIII] Corollary 3.12, in particular how the identifications across the Θ-link may be made; Mochizuki's position is set out in [EssLgc]. This guide describes the reconstruction machinery that both sides of that discussion take as input, and adjudicates nothing.

What we simplified. We wrote "the" fundamental group where basepoints and outer isomorphisms matter; we ignored orbicurve technicalities, the tempered versions at bad primes [AbsTopIII, Rem 1.9.1], and the minor corrections to [AbsAnab, Lemma 1.3.9] and [AbsTopI, Lemma 4.5] recorded in [IUTchI, Rem 1.2.2].

次回予告 · NEXT EPISODE

The groups have crossed the wall. The boxes are refilled. But the Θ-pilot is still on the wrong side, and the only thing everyone agrees on is a bento box! Can the log-Kummer correspondence, three indeterminacies and one very large log-shell describe it from over here? Next time, in the companion guide: "The Multiradial Algorithm". Don't miss it!

おまけ · Omake

Why did the profinite group refuse to go to the karaoke box? It only knew the songs up to isomorphism. Why does the tripod $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ always win at musical chairs? There are only three chairs, and Belyi showed everyone else is sitting on them anyway. What did the cyclotome say to the Kummer class? "Stay in tune, or you'll be f^{-1} by the bridge." And why do anabelian geometers never get lost on the Yamanote line? Because a loop determines the city.

★ References 参考文献

- [AbsAnab] S. Mochizuki, *The Absolute Anabelian Geometry of Hyperbolic Curves*, in Galois Theory and Modular Forms, Kluwer (2004). `pdf/Absolute Anabelian Geometry.pdf`. Prop 1.2.1: invariants of G_k.
- [AbsTopI–III] S. Mochizuki, *Topics in Absolute Anabelian Geometry I: Generalities; II: Decomposition Groups and Endomorphisms; III: Global Reconstruction Algorithms*, J. Math. Sci. Univ. Tokyo 19 (2012), 20 (2013), 22 (2015). `pdf/Topics in Absolute Anabelian Geometry I-III.pdf`. II §3: Belyi and elliptic cuspidalization; III §1: Thm 1.9, Cor 1.10, Thm 1.11; III §2: Aut-holomorphic spaces; III §3–5: log-Frobenius compatibility and log-shells.
- [Alien] S. Mochizuki, *The Mathematics of Mutually Alien Copies: from Gaussian Integrals to Inter-universal Teichmüller Theory*, RIMS Kôkyûroku Bessatsu B84 (2021). `pdf/Alien Copies, Gaussians, and Inter-universal Teichmuller Theory.pdf`. §2.6–2.13: transport; §3.4: cyclotomic rigidity, Fig 3.10.
- [EssLgc] S. Mochizuki, *On the Essential Logical Structure of Inter-universal Teichmüller Theory in Terms of Logical AND "∧"/Logical OR "∨" Relations*. `pdf/Essential Logical Structure of Inter-universal Teichmuller Theory.pdf`.
- [EtTh] S. Mochizuki, *The Étale Theta Function and its Frobenioid-theoretic Manifestations*, Publ. RIMS 45 (2009). `pdf/The Etale Theta Function and its Frobenioid-theoretic Manifestations.pdf`.
- [Gateway] S. Mochizuki, *Inter-universal Teichmüller Theory as an Anabelian Gateway to Diophantine Geometry and Analytic Number Theory*, Oberwolfach report (2023). `pdf/IUT as an Anabelian Gateway (MFO-RIMS23 Oberwolfach`

Report).pdf.

- [GCpLF] S. Mochizuki, *A Version of the Grothendieck Conjecture for p-adic Local Fields*, Internat. J. Math. 8 (1997). pdf/A Version of the Grothendieck Conjecture for p-adic Local Fields.pdf.
- [Hoshi19] Y. Hoshi, *Mono-anabelian Reconstruction of Number Fields*, RIMS Kôkyûroku Bessatsu B76 (2019).
- [Hoshi21] Y. Hoshi, *Introduction to Mono-anabelian Geometry*, Publ. Math. Besançon (2021), 5–44.
- [IUTchI–IV] S. Mochizuki, *Inter-universal Teichmüller Theory I–IV*, Publ. RIMS 57 (2021). pdf/Inter-universal Teichmuller Theory I-IV.pdf.
- [LocAn] S. Mochizuki, *The Local Pro-p Anabelian Geometry of Curves*, Invent. Math. 138 (1999). pdf/The Local Pro-p Anabelian Geometry of Curves.pdf.
- [pGC] S. Mochizuki, *The Profinite Grothendieck Conjecture for Closed Hyperbolic Curves over Number Fields*, J. Math. Sci. Univ. Tokyo 3 (1996). pdf/The Profinite Grothendieck Conjecture for Closed Hyperbolic Curves over Number Fields.pdf.
- [NSW] J. Neukirch, A. Schmidt, K. Wingberg, *Cohomology of Number Fields*, Springer.
- G. V. Belyi, *On Galois extensions of a maximal cyclotomic field*, Izv. Akad. Nauk SSSR 43 (1979); K. Uchida, *Isomorphisms of Galois groups of algebraic function fields*, Ann. of Math. 106 (1977); A. Tamagawa, *The Grothendieck conjecture for affine curves*, Compositio Math. 109 (1997).
- [Toolkit] This repository: src/ (iut run TransportMonoAnabelian, iut run ReconstructCoverings, iut ids), and scripts/anabelian/build.js, which computes the worked examples and builds this PDF.

Every show, game and character referred to above belongs to its creators and is mentioned only by description, with affection. Musubi is original. ありがとうございました！